

Slack Water

Tidal Prediction Lead · Sarn Barrage, twelve days before the springs

12 DAYS · 40 STOPS · GRADE 12 · SLACKWATER

THE OPENING CARD

A storm surge is coming in on the biggest tides of the year, half a metre on top of them. Six gates at Sarn Barrage are all that hold the water off four hundred hectares of marsh. You are the tidal prediction lead, and the gate programme they move on goes out over your name. You have twelve days, and no level goes on it unless you can show how you got it. Ninety graziers move stock off that marsh on one number you signed.

The board and the gauge

BEFORE THE DAY — TRIAL

Walk the barrage before the springs

Five buildings, a road across six gates and a pontoon on the west side: walk it once at low water so that you know the ground.

PLAN CARD 55W

Monday, twelve days before the springs. The gate programme is due. It owes a level for each tide, a bound on how wrong that level can be, and a line saying what neither covers. It rests on a tide nobody here has ever checked against a gauge. Today you break that tide into its parts.

BRIEFING 15W

Twelve working days to the equinoctial springs, and the gate programme has not been written.

WHAT YOU DECIDE 14W

Establish what the prediction is made of, and what the barrage does with it.

1.1 Two terms and a fortnight

PREDICTION OFFICE · A ROOM · DERIVE

SCENE 53W

Calloway, the chief tidal analyst, has the two biggest constituents on the board. One is a steady wave from the moon, one from the sun. This month both swing to the same height — the same amplitude. He wants the time from one spring tide to the next, using only these two waves.

GUIDE 93W

You build the working one line at a time. At each step pick the expression that the line above actually gives you. Every wrong branch here is real math. It is not a typo, so do not hunt for one. Two constituents added together can be rewritten as a product. That is the product-to-sum identity — the rule that turns "wave plus wave" into "wave times wave." One factor is fast. One is slow. That slow factor is the envelope. It decides how often the two waves swing back into step: the beat.

ASK 16W

Using the two constituents' speeds, how long is it from one spring tide to the next?

BEHIND THE BUTTON 183W

Why a product rather than a sum. Added together, two cosines of nearly equal speed give a wave. Its height rises and falls over days. Written as a product, that is explicit. One factor runs at the average speed. One runs at half the difference. The second factor is the envelope the first one runs inside.

Where the factor of two comes from at the end. The springs happen whenever the envelope is at its largest. The envelope is the size of the slow cosine rather than the cosine itself. A cosine reaches its largest size twice per period, once positive and once negative. So the interval between springs is half the period of the slow factor.

Why the amplitudes were made equal for this exercise. When the two amplitudes differ, the envelope never reaches zero. The arithmetic then acquires a term that carries no teaching. The real station has the moon term at 3.4 m and the sun term at 1.1 m. The interval between springs comes out the same. It depends on the two speeds and not on the two heights.

TAKES AS READ

a cosine of angular speed ω has period $360^\circ \div \omega$ · two terms are in step when their arguments differ by a whole number of turns

VERDICT 80W

The interval comes out to 14.7 days. Two constituents of nearly equal speed, added together. That's the same as multiplying a fast wave by a slow one — the product-to-sum identity again. The slow factor's speed is half the difference of the two speeds: 0.51 degrees an hour here. So the slow wave's period is 706 hours. The springs arrive at every peak of that envelope's size, and that happens twice each period. Half of 706 hours is a fortnight.

TAKEAWAY 20W

A beat's interval comes from the difference between two speeds. Never from either speed alone, and never from the amplitudes.

1.2 What the gates will see at two o'clock

SLUICE CONTROL · A PERSON · BALLPARK

SCENE 48W

Oyelaran, who runs the gates, wants the head across them at 14:00 before she signs the afternoon programme. The impoundment is being held at 4.30 m. Outside, the prediction is the mean level with two terms on top of it. Each one is already evaluated for that hour.

GUIDE 71W

Five numbers are on the table, and only three belong in the sum. Two of them are traps. They are the constituents' own amplitudes — the biggest number a term can ever reach, not what it is giving right now. Head is the gap between the level outside the gates and the level held inside. Find the outside level at 14:00, then subtract the impoundment level, and that is the head.

ASK 9W

What is the head across the gates at 14:00?

TAKES AS READ

a constituent contributes its amplitude only at the moment its cosine is one

VERDICT 86W

The head is 3.59 m. A tide term acts like a cosine wave. It only gives its full amplitude at the one moment its wave peaks. At any other hour it gives less, and on the far side of its cycle it takes level away instead. At 14:00 the sun term is negative, and the moon term is at about seven tenths of its amplitude. Head is the outside level minus the level held inside, both measured from the same datum — the same starting height.

TAKEAWAY 20W

An amplitude is the most a term can ever contribute. What it is contributing right now is a different number.

1.3 A clock and a position

SURVEY & DRIFT · SURVEY PONTOON · A ROOM · CHOICE

SCENE 45W

Owen Mbeki, the drifter technician, has float 7 on the screen at the pontoon. It went out on the ebb — the tide flowing out — ten days ago. It has sent back a position every ten minutes since. Nothing else is riding on it.

GUIDE 98W

A float like float 7 is what the course calls a Lagrangian drifter. It moves with the water instead of sitting fixed to the seabed. All four options here are things somebody might want from it. For each, ask whether a column of times and a column of positions could contain it directly. Sailors call these fixes. Then ask whether it would instead have to be worked out from how those columns change. What a record holds and what you can work out from it are different questions. Only one of the four sits directly in the record.

ASK 12W

What does float 7's record contain directly, before anything gets worked out?

TAKES AS READ

a position is two coordinates, and a track is those coordinates against time

VERDICT 89W

The record is where the float was and when it was there. Speed does not show up in it. Speed is how fast those coordinates are changing. Coordinates are the two numbers that make up a position. Getting speed means comparing the columns against each other. The same is true for direction and for distance covered. A path written as two coordinates against time is what the course calls a parametric curve. Every question about how the float moved is a question about the rates of its two coordinates.

TAKEAWAY 19W

A path written as two coordinates against time carries speed and direction inside it. Neither is spelled out, though.

END OF THE DAY 19W

A predicted tide is a sum of terms, and every question about it is a question about the terms.

A path, not a shape

PLAN CARD 44W

Tuesday. Float 7 has been adrift for eleven days. Nobody has ever checked its numbers against the meters. Today you find the float's speed and its heading, then decide how much of the tide the office can trust from eight terms instead of thirty-seven.

BRIEFING 14W

Float 7 is still out, and eleven days of fixes have never been reduced.

WHAT YOU DECIDE 11W

Get speed and heading out of a record that contains neither.

2.1 Two rates make one speed

SURVEY & DRIFT · SURVEY PONTOON · A ROOM · DERIVE

SCENE 69W

Sunniva Palmer, the hydrographic surveyor, has fitted a position function for each coordinate — a formula that gives it at any time. That came from twelve hours of fixes. One gives the easting (how far east), the other the northing (how far north). Both are in metres, with time in hours from local low water, when the tide is lowest. She wants the speed at $t = 2.0$ h.

GUIDE 73W

Palmer builds the working one line at a time. Each line has to follow from the expression right above it. Differentiate each coordinate on its own — that turns one path into two separate rates. Only after that do the two rates combine. They combine as the two shorter sides of a right-angled triangle, not by simple addition. Watch the inner factor on the easting closely: its angle is $\pi t/6.2$, not just t .

ASK 11W

What speed was float 7 making at $t = 2.0$ hours?

BEHIND THE BUTTON 216W

Why each coordinate gets differentiated on its own. Position here is just two numbers that happen to share a clock — the easting and the northing. Nothing in the easting's expression shows up in the northing's. So each is differentiated by itself. The pair of results together is the velocity vector. That's the arrow made from both rates at once, pointing the way the float is actually moving.

Why speed throws away direction, but the vector itself doesn't. Velocity is a pair of numbers — the vector — and speed is just one. Squaring both parts, adding them, and taking the square root gives the length of that arrow. It carries no record of which way it points. That arrow does still point somewhere. It lies along the tangent, the exact direction the path is heading at that instant. Speed is what's wanted next, because the next step is accumulating distance along the path. A direction doesn't factor into a length.

What the inner factor is doing. The easting is a sine of $\pi t/6.2$, not of t . That's because the tide comes full circle every 12.4 hours. Differentiating it brings that $\pi/6.2$ out the front — a factor of 0.507 per hour. Forgetting it doubles the eastward rate to something this estuary has never actually recorded.

TAKES AS READ

each coordinate can be differentiated on its own with respect to time · a rate of change, and the derivative as a rate — taken as read

VERDICT 78W

The speed is about 1,540 metres an hour. Differentiating the easting needs the chain rule, because it's a sine of $\pi t/6.2$, not of t . That inner factor is 0.507 per hour. It keeps the eastward rate at a believable 1,501 metres an hour. The northing is a

polynomial. So its rate falls steadily, reaching 324 metres an hour at $t = 2$. Speed is the length of the vector those two rates make — squared, added, and rooted.

TAKEAWAY 23W

Differentiating a path means differentiating each of its coordinates on their own. Speed is the length of the pair those two rates make.

2.2 The slope of a track with no formula

CURRENTS ROOM · A PERSON · BALLPARK

SCENE 45W

Dr. Halina Vos, the currents scientist, wants the float's track drawn on the chart with the right slope where it passes meter three. It can then be laid over the rose from the same hour. The two rates at that instant are on the board.

GUIDE 66W

Four quantities are offered and two of them belong in the answer. The track has no formula in easting and northing. So its slope on the chart cannot be found by differentiating one against the other. What is available is the two rates, and their ratio is the slope. The two entries that give a speed rather than a rate are there to be left out.

ASK 13W

What is the slope of float 7's track at $t = 2.0$ hours?

TAKES AS READ

a slope on a chart is a change in northing divided by a change in easting · the chain rule, and a quantity that moves because another one does — taken as read

VERDICT 102W

The slope is 0.216, about one north for every five east. Neither coordinate is a function of the other, so there is nothing to differentiate directly. What there is instead is how fast each is changing. Dividing the northward rate by the eastward one gives the slope of the path on the chart. The speed cannot appear, because it was made by throwing the direction away. Where the eastward rate passes through zero, this ratio has no value. The track has a vertical tangent there, not a data error. A vertical tangent just means the slope points straight up for an instant.

TAKEAWAY 21W

Where a path is given by two rates, its slope is their ratio, and the speed plays no part in it.

2.3 Eight terms of thirty-seven

PREDICTION OFFICE · A ROOM · PROTOCOL

SCENE 41W

Wren Achterberg is the harmonic analyst here. She studies the tide as steady waves at fixed speeds. She has all thirty-seven constituents on the wall. Only eight go on the daily board; the other twenty-nine are recorded and never added in.

GUIDE 67W

Each of the four cases on the left is a sum that stops partway through. A partial sum. Match each one to what stopping there really does to the answer. Here is the one thing to watch: in some sums the later terms keep shrinking. In others they do not. Only the shrinking kind lets you stop and still say something honest about what you left out.

ASK 13W

Match each way of stopping a sum to what it leaves you holding.

TAKES AS READ

a partial sum is what a sum comes to after a stated number of terms

VERDICT 95W

A partial sum is a choice about where to stop adding. The four cases differ in what that choice leaves you able to say. Listing what got dropped turns an unknown into a bound — a known ceiling. Dropping it with no list leaves a number nobody can defend. Stopping once the total settles down tells you about the terms you checked. It says nothing about the ones you didn't. Worst of the four is adding in fitted order instead of size order. That's because the biggest missing term could be anywhere in that order.

TAKEAWAY 19W

Stopping a sum early is a choice. It has to come with a statement of what that choice cost.

END OF THE DAY 18W

A record of positions against time holds the speed and the heading, but neither is written in it.

How far it went, and how far away it is

PLAN CARD 48W

Wednesday. The mud report is due next week. It needs how far a bit of mud travels, not just where it ends up. Today you find the length of a float's path over two hours, compare it with its straight-line drift, and learn why a rate cannot wait.

BRIEFING 15W

The sediment budget needs the distance float 7 travelled, not the distance it ended up.

WHAT YOU DECIDE 12W

Separate the length of a path from the gap between its ends.

3.1 Two thousand metres in two hours

SURVEY & DRIFT · SURVEY PONTOON · A ROOM · DERIVE

SCENE 47W

The launch surveyed float 7 nonstop for its first two hours. Palmer has fitted a position function for each coordinate, written as powers of time. She wants the length of that path in metres. That's the arc length, measured along every twist of it, not straight across.

GUIDE 81W

Palmer builds the working one line at a time, picking up whichever expression the line above gives. Length is just speed added up over time — that's what an integral does here. So the two rates come first, then the speed they make. Then the integral of that speed between the two times. Don't guess that last integral. The expression under the root has the derivative of its own inside sitting right outside it. That's exactly what makes a substitution work.

ASK 15W

How long is the fitted path over the first two hours, measured along the path?

BEHIND THE BUTTON 207W

Why the arc length is an integral of speed. The arc length is the true length measured along the path. It isn't the straight gap between its ends. Over a very short stretch of time, a path looks just like a straight line. Its length is speed times that short stretch. Adding up all those little pieces is exactly what an integral does. This is the one place in the course where the thing being added up is a length. It isn't a total amount of something.

Why the substitution is the natural one to try. The speed here is $600t\sqrt{1+t^2}$, and the derivative of $1 + t^2$ is $2t$. The t sitting outside the root is exactly half of what a substitution needs. So letting $u = 1 + t^2$ clears the whole expression. It turns it into the square root of a single variable.

What this answer is actually for. Nkechi Ude, the estuary ecologist, is asking how far a mud particle has really travelled through the water. That is the arc length. The float went in at one point. The straight-line distance between there and where it is now is a different number entirely. That one gets answered at the mud lab this afternoon.

TAKES AS READ

the length of a path is the accumulation of speed over the time spent on it · the definite integral as an accumulation of a rate — taken as read

VERDICT 95W

The path is about 2,040 metres long. Each coordinate gets differentiated on its own. The two rates go into the speed as the two shorter sides of a right-angled triangle. That speed then has to be added up over the whole two hours. That's what makes the length an integral instead of a simple multiplication. The float isn't moving at a constant speed at any point in those two hours. The substitution works because the

derivative of what's inside the root is already sitting right outside it. That's true up to a factor of two.

TAKEAWAY 30W

An arc length is speed added up over time. So a track that turns and loops is far longer than the straight gap. That gap runs between its two ends.

3.2 Six kilometres in eighty-four

MUD LABORATORY · A PERSON · BALLPARK

SCENE 43W

Ude has the eleven-day record of float 7 on the bench. The total of its ten-minute legs — the distance travelled — is 84 kilometres. She wants a different number: how far the last fix sits from the first, in one straight line.

GUIDE 71W

Four numbers sit on the bench, and only two of them build the answer you need. Displacement is the straight-line distance from where float 7 started to where it ended. It depends only on the first fix and the last one. Nothing that happened in between adds anything to it. The two changes in position run at right angles to each other. That is what tells you how to combine them.

ASK 14W

How far is float 7's last fix from its first, in a straight line?

TAKES AS READ

a displacement is fixed by two positions and by nothing in between them · arc length of a parametric path — taken as read · vector-valued functions, and acceleration as the derivative of velocity — taken as read

VERDICT 90W

The displacement is about 6,180 metres, against 84 kilometres of travel. Twice a day, the tide carries a float up the estuary and back down again. Almost all of that travel cancels out. What is left over two weeks is a small drift. Both numbers are real, and they answer different questions. A mud particle sits in 84 kilometres of moving water. It ends up only six kilometres from where it began. Using the smaller number where the bigger one belongs would understate the transport by a factor of thirteen.

TAKEAWAY 18W

Distance travelled divided by displacement tells you how much of a fortnight's travel the tide simply cancelled out.

3.3 The rate does not wait

SLUICE CONTROL · A ROOM · CHOICE

SCENE 50W

Tomas Beke, the sluice engineer, has the gate table on the desk. It says what to do at the top of each half hour. Each row was worked out from the flow at the moment it was written. Nobody has checked a row since the impoundment started filling this morning.

GUIDE 90W

Beke's table gives one instruction per half-hour row, and every row was worked out from the flow at the instant it was written. Head is the gap between the water level outside and the level held inside, and head is what drives the flow through a gate. So the question to put to this table is a narrow one. What has to be true of the flow for one instant's value to stand for a whole half hour? Then ask which way the error runs if it is not true.

ASK 16W

What does the table assume by using the flow at the top of each half hour?

TAKES AS READ

a rate quoted at an instant is the rate at that instant only

VERDICT 88W

It assumes the flow held for the whole half hour. But flow through a part-open gate depends on the head, and the head closes as the impoundment fills. So the true flow keeps falling through the interval. Taking the value from the start overstates the water passed, and the error grows with the interval's length. This is the assumption behind stepping a differential equation — an equation linking a rate to its own amount — forward in flat pieces. Its cost can be worked out, not argued about.

TAKEAWAY 18W

Using a rate over an interval assumes the rate held for the whole interval. It usually did not.

3.4 Two terms and a fortnight — Review

PREDICTION OFFICE · A ROOM · CALLBACK · BALLPARK

SCENE 48W

Wren Achterberg, the harmonic analyst, has two more of the daily-sum terms on the board. They are the pair that rises once a day rather than twice. Their speeds are about a degree an hour apart. She wants the interval between one time they add and the next.

GUIDE 79W

Four quantities are on the board and three of them make the answer. The interval is not the period of either term, and neither term's own period will get you to it. What decides it is how long the faster of the two takes to gain a whole turn on the slower. So the whole turn goes over the gap between the two speeds. The entry that is already a period in hours is there to be left out.

ASK 15W

How long is it from one time the two diurnal terms add to the next?

BEHIND THE BUTTON 120W

Why the two amplitudes do not appear. Adding two cosines of nearly equal speed gives a fast cosine inside a slow one. The slow factor is built out of the difference of the speeds alone. The amplitudes decide how far the water moves when the two add. They decide nothing about how often that happens.

Why this pair matters at Sarn. The semidiurnal pair comes round every 14.7 days, which is the fortnight the springs are counted in. The diurnal pair comes round on its own cycle, a little under a fortnight. The two are not in step with each other. The largest tides of the year happen where both patterns arrive together, which is what an equinoctial spring is.

TAKES AS READ

a cosine of angular speed ω has period $360^\circ \div \omega$ · two terms add most strongly when their arguments differ by a whole number of turns

VERDICT 93W

The interval is about 328 hours, which is 13.7 days. Two terms of nearly the same speed drift in and out of step at the difference of those speeds. A whole turn divided by that difference is how long the drift takes. Here the gap is 1.098 degrees an hour, so the pair comes back into step every fortnight or so. The semidiurnal pair on the board did the same thing for the same reason and arrived at 14.7 days. Amplitudes set how big the swing is and never how often it comes.

TAKEAWAY 22W

Two terms come back into step at the difference of their speeds, whatever their sizes and whatever each one's own period is.

END OF THE DAY 13W

Length is accumulated along a path; displacement only looks at the two ends.

What the tail can be worth

BEFORE THE DAY — TRIAL

The shore track, and the wall at the end of it

The wall station is two hundred metres along the shore to the east, and the walkway out to the gauge at the wall head is covered for four hours either side of high water. Take the track once in daylight, so that the day you are asked to fetch a gauge record you already know how long it takes.

PLAN CARD 47W

Thursday, six days before the springs. The board and the gauge get compared as numbers today, not opinions. Today you put a ceiling on what the board leaves out. Then you learn what the water does while a gauge settles, and what a meter's reading really means.

BRIEFING 16W

Calloway wants the gate call made on the board; Oyelaran wants it made on the gauge.

WHAT YOU DECIDE 15W

Put a ceiling on what twenty-nine dropped constituents can do, and compare it with waiting.

4.1 What twenty-nine terms can be worth

PREDICTION OFFICE · A ROOM · DERIVE

SCENE 48W

Achterberg has the twenty-nine dropped constituents — the small waves left off the board. She has them listed by size. 0.041, 0.033, 0.028 and 0.022 metres, then twenty-five smaller ones adding to 0.066. Calloway wants to know the worst those twenty-nine could do to a predicted water level.

GUIDE 79W

You build the working one line at a time. The remainder is the sum of the terms you left out. Each of those terms is an amplitude, a wave's height. Each is that amplitude times a cosine of some angle. You don't know what that angle is at the hour that matters. So the useful step is the one that stops needing to know it at all. A cosine can never be bigger than one, no matter its angle.

ASK 14W

At any single hour, what is the most the twenty-nine dropped constituents could add?

BEHIND THE BUTTON 218W

Why a ceiling, not a guess. The twenty-nine dropped constituents each turn at their own speed. At any one hour, some of them add together and some cancel out. So the real sum is almost always much smaller than the worst case. But nobody knows in advance which hour will matter most. That is why the honest number to print is the ceiling. The largest it could possibly be. Not a guess at the likely value.

Where the triangle inequality comes in. The size of a sum is never bigger than the sum of the sizes of its parts. That is the whole trick used here. It is also why the answer doesn't depend on any of the twenty-nine timing angles. It works no matter what those angles turn out to be. That independence is exactly what makes the ceiling worth printing.

Why the comparison at the end is the real point of the day. A ceiling is only useful next to something else. Twenty centimeters either way is a lot of water in a shallow estuary. But it is almost nothing next to how far the level moves. That's true even over the forty minutes a stilling well takes to settle down. Knowing which of those two is bigger is what decides how the gates get programmed.

TAKES AS READ

a cosine is never larger than one in size, whatever its argument · the size of a sum is at most the sum of the sizes of its terms · alternating series, and the error after n terms — taken as read

VERDICT 91W

The tail cannot go past 0.19 metres. Each dropped term is an amplitude times a cosine. A cosine never gets bigger than one in size. So each term is capped by its own amplitude, no matter its timing. Add those caps up and you get a ceiling on the whole tail. Not a guess, a hard limit. Most hours, the twenty-nine terms partly

cancel and add only a few centimeters. What makes the ceiling worth having is that it holds at every hour. Nobody needs to know the timings at all.

TAKEAWAY 22W

A sum that stops early still has a ceiling on its error. That ceiling needs no knowledge of the dropped terms' timing.

4.2 Forty minutes of water

SLUICE CONTROL · A PERSON · BALLPARK

SCENE 38W

The stilling well at the outer face needs about forty minutes before Oyelaran will read a level off it as settled. On a spring ebb the level outside is falling at nine tenths of a metre an hour.

GUIDE 54W

Four numbers are offered, and only two of them make the answer. The trap is the time. A rate in metres per hour needs a time in hours to match it, and forty is written in minutes. The prediction's bound is on the list for comparison later. It is not part of this arithmetic.

ASK 12W

How much does the level outside move while the well is settling?

TAKES AS READ

a rate multiplied by a time gives the change over that time, in the rate's own units

VERDICT 73W

The level moves about 0.60 metres while the well settles. That is three times this morning's ceiling on the twenty-nine unused terms. That comparison turns the argument into something you can decide with numbers, not just a guess. On a fast ebb, waiting for a settled reading costs more level than the board's worst-case error. That does not settle the question for every day. It settles it for a day dropping this fast.

TAKEAWAY 19W

The cost of waiting for a measurement is measured in the same units as the error of not waiting.

4.3 Ten minutes of current, in one number

CURRENTS ROOM · A ROOM · CHOICE

SCENE 43W

Pieter Loots, the instrument technician, has meter three open on the bench. It reports one speed and one bearing every ten minutes. Inside, it is sampling several times a second. It is also doing something to those samples before anything is written down.

GUIDE 47W

All four options are things the number on the record could be. For each one, ask what it would take hundreds of samples to build it. Then ask which of them you could still recover afterward. Only the one reading is kept. The samples themselves are gone.

ASK 11W

What is the number meter three writes down every ten minutes?

TAKES AS READ

an average over an interval is the accumulated quantity divided by the length of the interval

VERDICT 82W

It is the mean — the average — over the ten minutes. The samples are added up, then divided by the length of the interval. That adding-up is called accumulation. It is what an average value really is. Anything that happened faster than ten minutes gets folded into that accumulation. It cannot be pulled back out. That matters when the record is compared with a float. The float is somewhere specific at each instant. The meter has already smoothed the instants together.

TAKEAWAY 18W

A reported reading is an accumulation divided by its own interval. What happened inside the interval is gone.

4.4 A clock and a position — Review

SURVEY & DRIFT · SURVEY PONTOON · A ROOM · CALLBACK · PROTOCOL

SCENE 48W

Owen Mbeki, the drifter technician, has float 9 out of the water and its card on the bench. On the card are a column of times and two columns of position, ten minutes apart, and nothing else. Four numbers are wanted from that card for the sediment annex.

GUIDE 72W

Four quantities wanted from the card, and four descriptions of the work each one takes. Pair them. The test every time is how much of the record has to be touched. It may be a single line, or one neighbouring pair of lines. It may be two rates taken together as a right-angled pair. Or it may be every interval in the card added up from the first line to the last.

ASK 16W

Match each number the annex wants to the work the card makes you do for it.

TAKES AS READ

a fix is one position at one recorded time · a rate is a change divided by the interval it happened over

VERDICT 83W

A parametric record holds the position outright. It holds everything else as a rate or as a sum of rates. Speed comes from differencing both columns over one interval and combining the two rates as a right-angled pair, which is the size of the velocity. Heading comes from the same two rates as a ratio, which throws the size away and keeps the direction. Length has to be accumulated interval by interval, because no single line of the card contains any of it.

TAKEAWAY 22W

How much of a track you have to touch is what tells you which kind of quantity you are asking it for.

END OF THE DAY 21W

A bound on what a sum left out is a number, and it can be compared with the cost of waiting.

Speed against bearing

PLAN CARD 46W

Friday. The mud report needs the ebb and the flood compared as numbers. Nobody has ever taken one off these curves before. Today you find the area one curve encloses, compare it with the other, and learn why a gauge keeps rising after the tide turns.

BRIEFING 16W

The sediment work needs the ebb and the flood compared, and they are drawn as roses.

WHAT YOU DECIDE 11W

Get a number out of a figure that has no axes.

5.1 Wedges round a figure

CURRENTS ROOM · A ROOM · DERIVE

SCENE 41W

Vos has the ebb rose fitted. The reach at each bearing is 1.4 plus 0.9 times the cosine of the bearing, in metres a second. That runs over the half turn the ebb occupies. She wants the area the figure encloses.

GUIDE 74W

You build the working a line at a time. There are no axes to integrate against here. So the piece to work with is a thin wedge at one bearing. It is nearly a triangle, with the reach as its two long sides and a very short base. Adding the wedges round the figure is the integral. The square in it is why the expansion has to be done before anything can be integrated.

ASK 13W

What area does the ebb rose enclose, over the half turn it occupies?

BEHIND THE BUTTON 213W

Why a wedge and not a strip. A rose has no top and bottom, so a vertical strip cuts it in places that mean nothing. A wedge of angle $d\theta$ at reach r has area very nearly $\frac{1}{2}r^2d\theta$. It is a triangle with base $r \cdot d\theta$ and height r , and the error in that approximation vanishes faster than the wedge does.

Why the square is the awkward part. The reach is a sum of two terms, so its square has three: a constant, a cosine, and a cosine squared. The last one has to be rewritten using the double-angle form before it can be integrated. It is also the term that carries the asymmetry between the two halves of the tide.

What the number is for. This is a polar area — the space closed in by a curve that is drawn by bearing and reach instead of by x and y . On its own it is an index rather than a physical quantity: its units are speed squared times an angle, not square metres. What makes it worth having is that the same construction, applied to the flood rose, gives a number in the same units. The difference between the two areas is the transport asymmetry the flats are being built out of.

TAKES AS READ

a thin wedge of angle $d\theta$ at reach r has area very nearly one half r squared $d\theta$ · the definite integral as an accumulation of a rate — taken as read · polar coordinates, and a curve given as r of θ — taken as read

VERDICT 88W

The area comes out at about 6.24. A wedge at bearing θ has area one half r squared $d\theta$. So the area of the whole figure is half the integral of the square of the reach. Squaring the reach gives three terms, and each is integrated on its own. The constant gives 1.96 times π . The plain cosine integrates to zero over a symmetric half turn. The cosine squared has to be rewritten as one half of one plus the double angle before it will integrate at all.

TAKEAWAY 22W

A polar area is half the integral of the square of the reach, and the square is what makes the expansion necessary.

5.2 The gap between two roses

MUD LABORATORY · A PERSON · BALLPARK

SCENE 39W

Ude has both figures on the bench, with their squared reaches already integrated. They are the ebb rose and the flood rose, over the same half turn, on the same axes. She wants the transport imbalance as one number.

GUIDE 71W

Four quantities sit on the bench, and three of them belong in your answer. The area between the two roses is the outer rose's area with the inner rose's area taken out. That means each rose's square gets integrated on its own, before any subtracting happens. One of the four numbers comes from subtracting the reaches first and squaring second. That is the mistake almost everyone makes, so watch for it.

ASK 16W

What is the area between the ebb rose and the flood rose over the half turn?

TAKES AS READ

an area between two curves is the outer area with the inner one removed · area of a polar region as half the integral of r squared — taken as read

VERDICT 92W

The area between them is about 2.19, which is a third of the ebb rose's own area. Each rose's area is half the integral of its own squared reach. So the area between the two roses is half the integral of the difference between the squares. Squaring the difference instead gives 1.17 — the right units, the wrong area, and no way to tell from the number alone. The imbalance stays positive over this whole half turn. That is why the eastern flats are gaining ground and the western ones are not.

TAKEAWAY 19W

The area between two roses comes from the difference between their squares, never from the square of their difference.

5.3 Twenty minutes past the turn

WALL STATION · A ROOM · CHOICE

SCENE 55W

Fell keeps two years of notebook entries in the hut at the wall head. The wall head is the far end of the wall, away from the gates. Out here, the water keeps rising for about twenty minutes. That is after the water at the gates has already turned. This happens on every big tide.

GUIDE 66W

The water at one end of the wall is still rising. The water at the other end has already turned. All four choices are ways this could happen. Check each one two ways. Would it happen on every big tide? Would it grow bigger than the first arrival — the first wave that reaches the wall head? Those two checks are what Fell's notebook really shows.

ASK 16W

What is causing the wall head to keep rising for twenty more minutes, every big tide?

TAKES AS READ

a disturbance takes time to travel the length of a wall and back

VERDICT 96W

A wave of water runs down the wall, bounces off the far end, and comes back. That is the only choice that happens every time and keeps building. Each returning wave is a smaller copy of the last one. The waves arrive in step. That is because the round trip takes about as long as the forcing period — one tide cycle. So the gauge is not reading one wave. It is adding up a running total of many. That sum is made of shrinking pieces, so it settles on one number instead of growing forever.

TAKEAWAY 23W

When something bounces back again and again, the returns add up into one running total. That total is what the gauge actually measures.

5.4 The rate does not wait — Review

SLUICE CONTROL · A ROOM · CALLBACK · BALLPARK

SCENE 47W

Tomas Beke, the sluice engineer, has one row of last Thursday's gate table on the desk. Beside it is the recorder trace for the same half hour. The row was written from the flow at the top of it. The trace says what the flow averaged instead.

GUIDE 78W

Four quantities are on the desk and three of them belong in the answer. The row assumed one flow held for the whole interval. The trace gives what the flow averaged across it. The gap between those two rates, run for the length of the interval, is the water the sheet claimed and the estuary did not supply. The impoundment level is on the desk because the row was written from it, and it plays no part here.

ASK 14W

How much water does one row of the gate table claim that never passed?

TAKES AS READ

a rate quoted at an instant is the rate at that instant only · a volume is a rate multiplied by the time that rate ran for

VERDICT 89W

About 11,900 cubic metres, out of one row. Flow through a part-open gate follows the head, and the head closes while the water is passing. So the rate at the top of a half hour is the largest rate of that half hour. Using it for the whole interval charges the sheet the difference. Every stepped calculation carries that error, and it has two handles. One is how fast the rate is changing, which the estuary decides. The other is how long the step is, which the table decides.

TAKEAWAY 22W

What a rate costs when it is used over an interval is its gap from the average rate, multiplied by the interval.

END OF THE DAY 19W

A curve given as a reach at each bearing encloses an area, and the area is a comparable number.

What comes back, and comes back again

PLAN CARD 50W

Monday. The wall head reads far higher than the first wave that reaches it. The wall's armour was sized for that first wave alone. Today you add up the whole chain of waves that return, decide what a shrinking sum can prove, and learn what one float is evidence of.

BRIEFING 13W

The wall head reads higher than the first arrival, on every spring tide.

WHAT YOU DECIDE 17W

Add up a chain of reflections, and say when such a chain has a total at all.

6.1 Nine tenths out of a third

WALL STATION · A ROOM · DERIVE

SCENE 53W

Marta Grieve, the structures and resonance engineer, has the numbers from the wall head. The first arrival is 0.34 metres. Each return after that is about 0.62 of the one before it. It arrives right in step with the next tide cycle. She wants to know what the whole chain adds up to.

GUIDE 85W

You will build this a line at a time. First, write what the n th return is worth. Then write the partial sum — the total of just the first few returns. Next, look at what those partial sums are doing as you add more terms. The key step is naming what kind of sum this is. A neat formula for the total — a closed form — only exists for that kind of sum. It exists only when the returned fraction is less than one.

ASK 14W

What does the whole chain of returns add up to at the wall head?

BEHIND THE BUTTON 231W

Why the returns arrive in step. Three hundred metres out and back along the wall. That takes about as long as the forcing period at this station — one tide cycle. So each returning wave lines up with the crest of the next wave, instead of against it. If the waves arrived against each other, they would cancel out. Then the wall head would read less than the first arrival, not more.

Why the sum settles on one number. Each return is the same fraction of the one before it. So the terms shrink at a steady rate — this is called shrinking geometrically. Because of that, the partial sums close in on one value instead of growing forever. What decides the final total is not how many returns there are. It is how much of each return survives the round trip. At a returned fraction of 0.62, the total is 2.6 times the first arrival. At 0.8, it would be five times as large.

What this means for the armour. The last forty metres of armour was sized for the first arrival. That is the wave anyone standing at the gates would measure. But the armour actually has to meet the total, the full sum of the whole chain. How much bigger the total is than the first arrival depends only on the returned fraction. It depends on nothing else.

TAKES AS READ

each return arrives in step with the next forcing rather than against it

VERDICT 92W

The total is about 0.89 metres, against a first arrival of 0.34. The n th return equals the first arrival times the returned fraction raised to the n th power. That is the pattern that makes this a geometric series. A geometric series like this settles on the first term over one minus the fraction. Here the fraction is 0.62, so the total is 2.6 times the first arrival. The condition matters as much as the formula. At a fraction of one, there's no real total, though the algebra would still give a number.

TAKEAWAY 24W

What decides the size of a geometric sum is the fraction that survives one round trip. It is not how many returns you count.

6.2 Small terms and no total

PREDICTION OFFICE · A PERSON · CHOICE

SCENE 41W

Achterberg has two lists on the wall. In both, the terms get steadily smaller and shrink to nothing. One sum settles on a final value; the other never does. Calloway wants to know how the office can tell which is which.

GUIDE 68W

Here are four claims about a sum whose terms shrink to zero. For each one, ask what it actually lets you write in a report. Here is the key difference. A test can rule a sum out, or it can rule one in. This one only does the first. Terms shrinking to zero is required for a sum to settle. It is nowhere near enough on its own.

ASK 14W

If the office knows the terms shrink to zero, what can it actually conclude?

TAKES AS READ

a sum can only settle if its terms eventually become small

VERDICT 81W

On its own, it proves nothing. If the terms don't shrink to zero, the sum definitely has no total. So this test can rule a sum out. It cannot rule one in. The terms $1/n$ shrink to zero, and their sum still grows without limit. The terms $1/n^2$ also shrink to zero, and their sum settles just fine. Both have terms vanishing to nothing. What separates them isn't the size of any one term. It's how fast the sizes fall away.

TAKEAWAY 21W

Terms shrinking to zero is required for a sum to settle on a total. It is not proof that one exists.

6.3 Nine floats back out of eleven

SURVEY & DRIFT · SURVEY PONTOON · A ROOM · CHOICE

SCENE 50W

Palmer has already signed two reports she wasn't happy with. Both of them treated a single float's track as the transport pattern for the whole estuary. The estuary is the tidal stretch where the river meets the sea. Float 7 is about to become the third, unless somebody says something.

GUIDE 102W

Palmer is not being fussy. She has signed this mistake twice and does not want a third. Two of the words on the list are worth having before you choose. The harmonic prediction is the tide worked out from the sun and moon's cycles. The exchange fraction is how much of the impoundment's water swaps out each tide. Now go down the options and ask of each one whether it treats the float as a measurement of what happened. Or as a model of what generally happens. Then ask what that difference costs a reader making a decision two years from now.

ASK 8W

What is float 7's track actually evidence of?

TAKES AS READ

a measurement describes the occasion it was taken on

VERDICT 93W

It is evidence of what the water did during those eleven days. That's worth a lot — it's the only direct measurement of transport this estuary has. What it can't do is stand in for the general pattern. Two weeks of tides has one run of winds and one spring-neap phase. That's where tides swing extra strong or extra weak. A prediction covers the general case and comes with a bound, a stated range of error. A measurement covers one case exactly. Reports go wrong when a measurement gets promoted into a prediction.

TAKEAWAY 19W

A measurement is evidence for its own occasion. It becomes a model only when someone says so in writing.

6.4 The slope of a track with no formula — Review

CURRENTS ROOM · A ROOM · CALLBACK · CHOICE

SCENE 42W

Dr. Halina Vos, the currents scientist, has meter two's record for the turn of the ebb. For ten minutes either side of slack the eastward rate runs down through zero. The northward rate is still going at three hundred metres an hour.

GUIDE 69W

All four options describe the chart at the same instant. Ask first which of the two rates sits underneath in the ratio that makes a chart slope. Then ask what a quotient does as that one closes on zero. Then ask separately what is actually misbehaving at that moment. Is it the water, the float or either rate, or only the number built out of the two of them?

ASK 14W

What is the track's chart slope doing as the eastward rate passes through zero?

TAKES AS READ

a slope on a chart is a change in northing divided by a change in easting · a ratio whose denominator vanishes has no finite value · the chain rule, and a quantity that moves because another one does — taken as read

VERDICT 85W

The chart slope is the northward rate over the eastward one. So it grows without limit as the eastward rate closes on zero, and the track stands vertical on the chart. Nothing is wrong with the float or with either meter at that instant. Both coordinates are moving smoothly and both rates are perfectly well defined. What has failed is the description. A chart slope asks how far north per metre east, and for those few minutes there are no metres east to divide by.

TAKEAWAY 20W

A path can be perfectly smooth at an instant where its slope on the chart has no value at all.

END OF THE DAY 23W

A chain of returns each a fixed fraction of the last has a finite total, and the total is not the first term.

The impoundment fills

PLAN CARD 52W

Tuesday, six days before the springs. The gate plan depends on how long it takes to fill behind the gates. That rate falls the whole time it works. Today you solve the fill equation, take one step of the storm forecast forward by hand, and learn what a tide truly washes out.

BRIEFING 19W

Beke wants a filling time he can defend, and a reason for half an hour rather than ten minutes.

WHAT YOU DECIDE 14W

Solve the filling equation, and price the interval the gate table is written to.

7.1 A rate that closes its own head

SLUICE CONTROL · A ROOM · DERIVE

SCENE 45W

Beke has the flow law for a part-open gate. The level inside rises at a constant times the square root of the head across it. Outside is held at 7.4 metres, the impoundment starts at 4.3, and the constant is 0.11 in metres and minutes.

GUIDE 100W

You build this line by line. The head across the gate is the gap between the outside level and the impoundment's own level, and it is what drives the rate. Put every h on one side of the equation and every t on the other side. That split is what makes this differential equation solvable at all. A differential equation like this one just ties a rate to the amount it is the rate of. Then integrate each side on its own. The starting level fixes the constant, and only after that is there anything to rearrange for a time.

ASK 13W

How long does the impoundment take to come up to the outside level?

BEHIND THE BUTTON 195W

Why the rate is a square root. Flow through an opening is driven by the head across it. The speed of that flow goes as the square root of the head. That is the same relation that makes a jet from a deep tank faster than one from a shallow one. So the impoundment comes up quickly at first and then more and more slowly.

Why this cannot be done by multiplying. The rate at the start is 0.11 times the root of 3.1, which is 0.194 metres a minute. At that rate the impoundment would be level in 16 minutes. The true answer is twice that, because the rate is falling the whole way and the starting value is the largest it ever takes.

Why the solution reaches the end instead of just approaching it. An exponential decay is a change that slows down forever and never quite stops, so it only approaches its target. This one is a square root instead, so the solved shape is a parabola in time. It meets the outside level exactly, at one finite minute. That is a real difference between two equations that otherwise look equally harmless.

TAKES AS READ

an equation whose variables can be put on opposite sides can be integrated side by side · the fundamental theorem, and the antiderivative it rests on — taken as read

VERDICT 86W

The impoundment takes about 32 minutes. Putting every h on one side and every t on the other leaves two integrals that can each be done. The h side integrates to minus twice the root of the head. The starting level fixes the constant. The level meets the outside level when that root reaches zero, which is a finite time rather than a limit. Multiplying the starting rate by anything would have given 16 minutes, because the starting rate is the fastest the gate ever runs.

TAKEAWAY 17W

Separating the variables turns a rate that depends on the answer into two integrals that do not.

7.2 Half an hour of tangent

PREDICTION OFFICE • A PERSON • BALLPARK

SCENE 46W

The surge forecast comes in as a rate, not a height. Right now the surge grows by 0.42 metres an hour. It decays at 0.15 times its own current height. It stands at 0.30 metres, and the program wants its value half an hour from now.

GUIDE 76W

Five numbers are offered, and only three of them belong in the step. The model's real rate is already on the list. It's worked out at the surge's present height. The raw growth number by itself is also on the list, as a trap. Using that instead means stepping forward on a rate the model doesn't actually have right now. And the step size has to be in hours, since the rate is measured per hour.

ASK 12W

After one half-hour step, what does the model give for the surge?

TAKES AS READ

a rate can be treated as constant over a short enough interval • separable differential equations — taken as read

VERDICT 82W

One step gives 0.49 metres. This method pretends the rate stays constant for the whole half hour, and it does not. The decay term grows as the surge grows. So the true rate keeps falling across that half hour, and the step overshoots. Cutting the interval in half roughly cuts that overshoot in half too. That's the real reason Beke keeps asking for smaller steps. The raw growth number alone is only the model's true rate when the surge is exactly zero.

TAKEAWAY 22W

A step forward equals the current value plus the current rate times the time interval. The interval is where the error hides.

7.3 A third of a third of a third

MUD LABORATORY · A ROOM · CASEBOOK

SCENE 40W

Ude has the dye results from March on the bench. A third of the impoundment's water is exchanged on every tide. The licence report claims the impoundment is clear after three tides. But the dye was still detectable after nine.

GUIDE 81W

Each thing Ude noticed on the left has one explanation on the right that fits it. What splits the four explanations is one question. When water leaves the impoundment on the ebb, is it fully replaced by new water? Or does some of that same water just come back on the next flood? That is the difference between a total that empties out and a total that keeps shrinking by the same fraction, tide after tide. It never actually hits zero.

ASK 10W

Match each observation to the explanation that accounts for it.

TAKES AS READ

what leaves an estuary on the ebb is partly what comes back on the flood

VERDICT 101W

All four explanations are the same arithmetic, seen from different angles. A tide that exchanges a third of the water leaves two thirds behind. So after n tides, the fraction left is two thirds multiplied by itself n times. The sequence runs 0.67, then 0.44, then 0.30, and about 0.026 after nine. Each tide removes a third of a smaller amount than the tide before it, so the kilograms removed fall too. Some of the water that left is still sitting just outside the gates when the next flood begins. That is why the fraction exchanged is not the fraction cleared.

TAKEAWAY 20W

Removing a fixed fraction each cycle always leaves a fixed fraction behind — and that fraction never actually hits zero.

END OF THE DAY 16W

A rate that depends on what it has already done has to be solved, not multiplied.

The tail nobody measures

BEFORE THE DAY — FOLLOW

The core route, walked by the man who walks it

Cal Reddy, the sediment technician, takes fourteen cores in the same order on the same state of the tide, and the route is not on any chart. Stay with him along it. Where he stops is where the flats will take a boot, and the two stations he cannot reach on foot are the two that go missing.

PLAN CARD 57W

Wednesday. The mud report needs the whole tail of a discharge that never quite ends. Only some such tails add up to a real number. Today you take that tail out to infinity, and sort four tails into those with a total and those with none. Then you learn why the wall's count is the same question.

BRIEFING 17W

The nutrient claim in the licence report is an integral to infinity, and nobody has taken it.

WHAT YOU DECIDE 13W

Decide when the far end of a long process is worth a number.

8.1 For ever, and forty-eight kilograms

MUD LABORATORY · A ROOM · DERIVE

SCENE 39W

Ude has fitted the decay curve. The dissolved load starts at 48 kilograms and falls by a fixed fraction on every tide, with a decay time of 6.2 tides. The report wants the total exposure over the whole decay.

GUIDE 97W

Build this working one line at a time, and check each line against the rule it is allowed to use. This is an improper integral — an integral with no upper end. You cannot evaluate that directly at infinity. Instead, you evaluate it out to some finite end, and then let that end move out towards infinity. That one step — sending the end out — is what decides whether an answer exists at all. The antiderivative itself is the ordinary one, nothing special. What matters is what that antiderivative does as the upper end keeps growing.

ASK 11W

What is the total exposure over the whole decay, in kilogram-tides?

BEHIND THE BUTTON 245W

Why the upper end has to be a limit, not a number. Infinity cannot be plugged into a formula the way a number can. So the integral is defined a different way. Take its value at some finite upper end. Then ask what happens to that value as the end moves farther and farther out. For this decay, the piece of the answer that depends on the upper end shrinks to nothing as the end grows. That is why a finite answer survives.

Why this tail — the part that runs on forever — ends up with a total, when some tails do not. A quantity that falls by a fixed fraction every tide falls faster than any power of time. That is fast enough for its area to add up to a finite number. A quantity that falls like one divided by time does not fall fast enough. Its integral grows like a logarithm, and a logarithm eventually passes every value, no matter how large. That is the same split that decides whether an infinite sum of shrinking terms adds up to a total or not.

What the units mean. A load measured in kilograms, added up over tides, gives kilogram-tides. That is a measure of exposure, not a measure of mass. It is the right quantity for the nutrient claim. What harms the flats is how much dye was present, and for how long. It is not just how much arrived in total.

TAKES AS READ

an integral with no upper end means the limit of the integral as the end moves out · the integral test, and comparison between series — taken as read

VERDICT 111W

The total exposure is about 298 kilogram-tides. First, the integral is taken out to some finite upper end. That gives the decay time, times the starting load, times one minus a term evaluated at that end. Then the end is sent out toward infinity. Because

the load falls by a fixed fraction every tide, that end term shrinks to nothing. So the whole tail adds up to a finite amount. What is left is the starting load times the decay time — a clean and useful result. The exposure of a decaying discharge equals what it would be if the discharge held its very first value for exactly one decay time.

TAKEAWAY 24W

An improper integral is a limit — and that limit is exactly where a tail either adds up to a total, or does not.

8.2 Four tails, two totals

PREDICTION OFFICE • A PERSON • PROTOCOL

SCENE 37W

Achterberg has four models on the wall. Each gives the size of the n th tide's contribution after a discharge. In all four, the contributions get smaller. Only two of the four add up to a final number.

GUIDE 70W

Match each tail on the left to what its total actually does. Every one of these has terms shrinking to nothing, so that alone won't tell them apart. What separates them is how fast the terms fall. One kind shrinks by a fixed fraction each time. The other shrinks by a fixed power of the tide number. For that kind, it matters whether that power is above or below one.

ASK 8W

Match each tail to what its total does.

TAKES AS READ

the size of the terms decides the total only through how fast they fall • the n th-term test, and what it can and cannot settle — taken as read

VERDICT 94W

A fixed fraction each time makes a geometric series, and it does settle. Here at $1 \div (1 - 0.6)$. The other three are powers of n , called a p -series. Its boundary sits right at power one. $1/n$ has no total, though it only grows as slowly as a logarithm. 1 over the square root of n is even worse. Its terms fall away more slowly still. $1/n^2$ does settle on a total. All four have terms shrinking to nothing. That is exactly why the size of one term alone never settles this question.

TAKEAWAY 22W

Whether an endless sum settles on a total comes down to how fast its terms fall. Not just that they get small.

8.3 Two hundred and forty units

WALL STATION · A ROOM · CHOICE

SCENE 43W

Grieve has the armour survey on the table in the hut. There are two hundred and forty armour units in a row. Each one takes a share of the wave energy that reaches it. The share gets steadily smaller toward the wall head.

GUIDE 62W

All four choices are ways to add up a run of two hundred and forty terms. For each one, ask: does it need every single term measured? Then ask: what would it do if the wall were made longer? The winning method is the one that compares the terms to something. You already know how to find the area under that something.

ASK 34W

Grieve needs to put a safe upper limit — a bound — on the total share of the units. These are units she has not surveyed. How can she do that without measuring them?

TAKES AS READ

a sum of terms that fall steadily can be compared with something. That something is the area under a curve through them · the harmonic series and p-series — taken as read

VERDICT 88W

The terms fall steadily along the wall. So the total of the unsurveyed units is capped by the area under a curve. That curve is drawn through them. That area is an ordinary integral. This trick is called the integral test, and here it does real engineering work. It turns an unmeasurable sum into an integral you can compute. That gives a safe limit, not a guess. Scaling up would assume the far units act like the near ones. The steady drop in the data says they don't.

TAKEAWAY 23W

When terms fall steadily, capping their sum with an integral tells you the leftover part is safe. That's true even without measuring it.

END OF THE DAY 23W

An integral or a sum that runs for ever may still be a finite number, and which it is has to be shown.

The residual has a shape

PLAN CARD 52W

Thursday, three days before the springs. What this office has called noise for nine years turns out to have a shape. Today you expand a rule to find where that shape comes from. Then you pick which fit to carry forward, and settle the order the gate plan must be built in.

BRIEFING 15W

The eight-constituent fit leaves something behind, and it is not the same at every station.

WHAT YOU DECIDE 18W

Find out where the extra term in the record comes from, and whether the fit has absorbed it.

9.1 Where the second term goes

PREDICTION OFFICE · A ROOM · DERIVE

SCENE 46W

Calloway has the discharge law on the board. Flow through the gates rises as the head to the power three-halves. Mean head is 2.1 metres, and the tide moves it by 0.85 metres either way. He wants that law rewritten as a polynomial in that movement.

GUIDE 69W

You build the working one line at a time. An expansion around a point is built from the value there, plus its derivatives. Each one is divided by its matching factorial. Take the derivatives first. Then assemble them into a second-order expansion. Only after that do you put the tidal movement in. The interesting part happens to the squared term once that movement is a cosine — a constituent.

ASK 16W

When the head moves like a cosine, what does the discharge law's second-order term turn into?

BEHIND THE BUTTON 191W

Why expand the law at all. The power three-halves is not a polynomial. So feeding a sum of constituents into it does not return a sum of constituents. Expanding turns the law into powers of the movement instead. And powers of cosines are something a harmonic prediction can actually hold. A harmonic prediction is one built from steady waves.

What the squared term does. A cosine squared equals one-half of (one plus the cosine of twice the angle). So the second-order term of this expansion splits into two parts. A constant, and a wave at twice the tidal speed. That second part is a constituent the fit never measured. It isn't made by the moon at all. It is made by the shallow water itself.

Why stopping at first order isn't enough. To first order, the flow just scales with the head's movement — which is symmetric. That means exactly as much extra flow arrives on the incoming tide. The same amount is missing on the outgoing one. But every real asymmetry the estuary shows lives in the second-order term and beyond. That includes the one seen in the two roses.

TAKES AS READ

a polynomial in a small movement can stand in for a smooth law near a point

VERDICT 108W

The second-order term turns into a constant plus an overtide. That's a wave at twice the tidal speed. The expansion's coefficients are the derivatives at mean head, each divided by its factorial. Three-halves over H for the first one, three-eighths over H squared for the second. Put a cosine into that squared term, use the double-angle rule. It splits into two pieces. A steady part, and a part swinging twice as fast. Here the movement is 0.85 metres on a mean head of 2.1. So the overtide comes out to

about three percent of the mean discharge. That's a size that shows up clearly in a nine-year record.

TAKEAWAY 22W

Expanding a curved law to second order creates a wave at twice the driving speed. No first-order sum could ever contain that.

9.2 Three fits, one to carry forward

CURRENTS ROOM • A ROOM • RESIDUAL

SCENE 46W

Six stations, each with two weeks of tide readings. At each one, Vos has the leftover — the predicted high water minus the one actually measured. She has fitted that same fortnight three separate ways. Only one of the three fits goes into the springs prediction.

GUIDE 82W

Each stick is one station, drawn where it sits in the estuary. Up is a prediction that came in low, and down one that came in high. Look at every field before you choose. The summary number at the top of each is the typical size of what is left over. It is not the only thing worth reading. A field where the leftovers change sign as you move up the estuary is telling you about something the fit does not contain.

ASK 17W

Look at each field before you choose. Which of these fits should go into the springs prediction?

TAKES AS READ

a fitted parameter will absorb whatever pattern is put in front of it • alternating series, and the error after n terms — taken as read

VERDICT 93W

Two of the three leave a pattern: low seaward, high up-estuary, at every station. That is not scatter. It is a term the model does not contain. Fitting a free offset at each station soaks it up and improves the summary number, while leaving the cause exactly where it was. The offsets will then be wrong by a different amount at the springs, when the movement is larger and the squared term matters more. The shallow-water fit scores worse and its leftovers change sign at random, which is what a complete model produces.

TAKEAWAY 21W

A fit that scores better by absorbing a pattern into a free parameter leaves the cause in place and hides it.

9.3 What to write down first

SLUICE CONTROL · A PERSON · SEQUENCE

SCENE 44W

Beke has four things to do before the programme can be issued, and no agreed order for them. Two of them can be undone in an afternoon. Two of them cannot be undone at all once the sheet has gone to the gate crews.

GUIDE 61W

Four steps, and this is not about scheduling by time. All four of them could be started this morning. Put them in order of what each one commits you to, freest first. What decides the order is which steps you could still change your mind about afterward. One of the four is the one that cannot be taken back at all.

ASK 11W

Order the four steps by what each one commits you to.

TAKES AS READ

a step that can be revised costs less than one that cannot

VERDICT 90W

Recomputing costs an afternoon of somebody's time and can be thrown away. Printing the bound changes nothing about the decision but makes it reviewable, so it is nearly free. Agreeing a closing level in writing binds two people and can still be revisited. Issuing the sheet and standing up a marsh warning reaches people outside this building. The marsh graziers move stock on it, and that cannot be recalled by saying so. The order is about what each step takes away from you, and only the last one is irreversible.

TAKEAWAY 20W

Order work by what it closes off, and put the step that cannot be undone last, however ready it seems.

END OF THE DAY 17W

A residual with a pattern in it is a term the model does not contain, not scatter.

An honest bound on the wrong quantity

PLAN CARD 49W

Friday. A storm is bringing half a metre of extra water overnight. Neither of the office's two limits was built for storms. Today you set a limit on a new kind of error. You decide what that limit leaves out. Then you learn how fast a current can change.

BRIEFING 13W

The surge forecast has arrived, and it is not in the constituent list.

WHAT YOU DECIDE 13W

Bound what the expansion left out, and find out what neither bound covers.

10.1 One more derivative, over six

PREDICTION OFFICE · A ROOM · DERIVE

SCENE 43W

The expansion is about to be used in the springs prediction. There, the head moves 0.85 metres around a mean of 2.1. Calloway won't sign off on it without a ceiling. He wants to know what stopping at the squared term actually costs.

GUIDE 79W

You build the working one line at a time. The ceiling is built from the next derivative. That's the one right after the last one you kept. Take that derivative's largest size anywhere between the expansion point and the point you are using. Multiply that by the step, raised to the matching power, then divide by the matching factorial. Each of those three pieces is a place the math can go wrong. It can still spit out a number.

ASK 13W

Across the tidal movement, how large can the second-order expansion's error possibly get?

BEHIND THE BUTTON 162W

Why the next derivative, not the next term. The expansion's next term uses the derivative right at the expansion point. The ceiling instead uses the largest that same derivative gets anywhere across the whole interval. That's exactly what turns a mere estimate into a true ceiling.

Where that largest value actually sits. The third derivative here carries a negative power of the head. So it is biggest where the head is smallest. That's at the low-water end of the movement. Using mean head instead gives a number about a fifth too small. Nothing in the arithmetic will warn you.

What the ceiling is actually worth. It comes out to about half a percent of the mean discharge. That's comfortably inside what the gate program can absorb. That is the honest result, and it is a ceiling on the expansion alone. It says nothing about anything the expansion was never given in the first place. That's the subject of the next call today.

TAKES AS READ

a bound uses the largest the next derivative reaches on the interval. Not its value at one point · alternating series, and the error after n terms — taken as read

VERDICT 99W

The ceiling comes out to about 0.4 percent. What's left over after a second-order expansion is capped by the largest size. The third derivative reaches that size somewhere on the interval. Multiply that by the step cubed, then divide by three-factorial. Here the third derivative goes as the head to the power minus three-halves. So it is biggest at the smallest head on the interval, not in the middle. The middle value would understate the ceiling. Half a percent is small enough that the

expansion is safe to use. That answers what Calloway wanted, though not today's real question.

TAKEAWAY 21W

A Lagrange bound is a ceiling on an expansion that stopped early. It says nothing about terms the model never had.

10.2 Nineteen centimetres of what

SLUICE CONTROL · A PERSON · CHOICE

SCENE 45W

Oyelaran has both of Calloway's bounds in front of her and the surge forecast beside them. The forecast is half a metre of wind-driven water, arriving on the largest range of the year. It came from the weather service rather than from the constituent list.

GUIDE 56W

Four readings of what a 0.19-metre bound covers. Ask of each what the bound was computed from. A ceiling can only be a ceiling on the terms that went into it. The distinction that matters is between two kinds of term. One was dropped from the sum. The other was never in the sum at all.

ASK 10W

What does the 0.19-metre bound on the dropped tail cover?

TAKES AS READ

a bound is computed from the terms of a particular model

VERDICT 105W

It covers the twenty-nine constituents that were fitted and then left out of the daily sum. That is all it was built from: their amplitudes, bounded one by one. A surge is not one of the thirty-seven constituents, because it does not repeat on any tidal cycle. It comes from wind and pressure instead. A harmonic fit only adds up repeating tide terms, so a prediction built that way has no term for a surge at all. Both of Calloway's bounds are correct. Neither one is a bound on the level, though, because the level tomorrow has a term in it that neither model has.

TAKEAWAY 19W

A bound is a statement about a model, so anything outside the model is outside the bound as well.

10.3 The rate of a rate

CURRENTS ROOM · A ROOM · CHOICE

SCENE 47W

Vos has meter three's fortnight on the screen as two components instead of as speed and bearing. One component is east-west, the other is north-south. The mooring designers have asked her for how fast the current changes. She wants the question put properly before she answers it.

GUIDE 79W

Vos will not answer until the question is put properly. She is right to wait. How fast a current changes can mean more than one thing. Ask of each option what it is the derivative of. Then ask whether it can be got from the two components on her screen. A current has a direction as well as a size. So a change in it can be a change in size, a change in direction, or both at once.

ASK 13W

What should Vos give the mooring designers for how fast the current changes?

TAKES AS READ

a velocity is two components, each a function of time

VERDICT 80W

Differentiating each component gives the acceleration. That is the quantity that includes turning. A current can hold a steady speed and swing through ninety degrees in an hour. The mooring feels that as a change of load direction. The derivative of the speed on its own is blind to it. Speed was made by throwing the direction away. Which of the two the designers want is a real question. The answer that contains both is the pair of component derivatives.

TAKEAWAY 17W

Differentiating a velocity component by component gives an acceleration that includes turning as well as speeding up.

END OF THE DAY 18W

A bound covers what the model it came from contains, and a surge is not in the model.

Two integrals that will not come out

PLAN CARD 46W

Monday, two days before the springs. The report must say when in the tide the water left, not just how much. Today you find that time as an average. You split a hard fraction into simple parts. Then you decide when a small fix can stop.

BRIEFING 18W

The generation report wants when the water went, and the silt model will not integrate as it stands.

WHAT YOU DECIDE 15W

Get a moment and a growth law out of integrals that need a technique each.

11.1 When in the ebb the water went

SLUICE CONTROL · A ROOM · DERIVE

SCENE 42W

Beke has the release profile for the first quarter of an ebb. The flow starts at its largest and falls away as a cosine, reaching nothing after 3.1 hours. The report wants the mean time of the release, weighted by the flow.

GUIDE 61W

You build this line by line. The integral you start with has a time multiplying a cosine, and neither piece is the antiderivative of anything simple. So the integral has to be traded for a different one. Which factor you choose to differentiate decides whether what comes back is easier or harder. Only one of the two choices actually makes progress.

ASK 14W

What is the flow-weighted mean time of the release, in hours into the ebb?

BEHIND THE BUTTON 157W

Why look at the moment and not just the total. The total release tells you how much water came out. The moment tells you when. Two releases with the same total can put their water an hour apart in the ebb. Both the turbines and the marsh care which.

Why differentiate the time. Differentiating the time gives one, which removes it from the problem. Differentiating the cosine leaves a time multiplying a sine, which is exactly as hard as what you started with. That is the classic sign that the two factors were chosen the wrong way round.

What the answer means. The flow-weighted mean time comes out at about 1.13 hours into a 3.1-hour quarter, rather than at the middle. That is because the flow is largest at the start, so the water is weighted towards the beginning of the release. A report that assumed the midpoint would put the whole release half an hour late.

TAKES AS READ

a mean weighted by a rate is the moment of that rate divided by its total · the fundamental theorem, and the antiderivative it rests on — taken as read

VERDICT 87W

The mean time is about 1.13 hours, not the 1.55 hours the midpoint would give. Parts turns the integral of a time times a cosine into a boundary term plus the integral of a sine, which comes out. Choosing to differentiate the time is what makes that happen: it becomes one and leaves. The moment divided by the total release is the mean. It sits early in the quarter because the flow is largest at the start, which is the whole reason the quantity was asked for.

TAKEAWAY 19W

Parts trades one integral for another, and the trade is only worth making if the factor you differentiate disappears.

11.2 One fraction, two denominators

MUD LABORATORY · A PERSON · CHOICE

SCENE 45W

Ude's silt model says the flats fill fastest when plenty of silt is already there, and plenty of room is left. Separating the variables leaves one fraction, one over the product of two brackets, on the left. Nothing in the bench's integral tables fits it.

GUIDE 96W

Ude has one fraction with two brackets multiplied underneath it. Nothing in the bench's tables fits that shape. What she can do is partial fractions. Split that one fraction into a sum of two simpler ones, each with a single bracket underneath. Each of those does have a standard antiderivative. That means a function the tables already know how to undo. Ask of each claim what the split makes possible that was not possible before. Then ask a second question. Does the split change the value of the integral, or only the way it is written?

ASK 11W

What does splitting the fraction into two simpler ones actually achieve?

TAKES AS READ

a fraction with a factored denominator can be rewritten as a sum of simpler fractions
• the fundamental theorem, and the antiderivative it rests on — taken as read

VERDICT 83W

Splitting turns something with no standard antiderivative into two pieces that each have one. Each piece integrates to a logarithm. The sum of the two simpler fractions is identically equal to the original fraction. So nothing about its value has changed — only its form. The same integral gets taken, just by a route that actually exists. Combined with separating the variables earlier, this is what produces the S-shaped curve the flats actually follow. It fills fastest when they are about half full.

TAKEAWAY 16W

Splitting a fraction changes nothing about its value and everything about whether it can be integrated.

11.3 A bracket round a refinement

PREDICTION OFFICE · A ROOM · CHOICE

SCENE 35W

Achterberg is refining the springs prediction one correction at a time. Each correction flips sign from the last, and each is smaller. She has made her fourth correction; the fifth, still unused, is 0.011 metres.

GUIDE 52W

Here are four claims about a sequence of corrections that flip sign and shrink each time. For each claim, ask what the next unused correction tells you. It's about where the true value really sits. Then ask one more thing: does the answer need any knowledge of the corrections after that one?

ASK 12W

After four corrections, what can Achterberg actually say about the true value?

TAKES AS READ

each correction is smaller in size than the one before it and has the opposite sign

VERDICT 99W

It lies within 0.011 metres of the total she already has. That's because the corrections alternate in sign and keep shrinking. Each partial sum overshoots the true value, and the next one undershoots it. So any two neighboring partial sums bracket the real answer between them. The width of that bracket is just the size of the first correction left out. The fifth one here. That's a true ceiling, not a guess, and it needs nothing about corrections beyond the fifth. It's also the cheapest ceiling in the whole course. The number is already sitting on the desk, unused.

TAKEAWAY 21W

Corrections can alternate in sign and keep shrinking. When they do, the true answer sits bracketed between two neighboring partial sums.

END OF THE DAY 17W

An integral that will not come out is usually an integral that has not been rearranged yet.

The gate programme

PLAN CARD 49W

Tuesday. The springs arrive tomorrow. Two signatures that have argued all fortnight both go on one sheet today. Today you check a rule still holds at tomorrow's peak. Then you decide how to spend what is left of the survey money, and settle what the east gate closes on.

BRIEFING 21W

The programme for day thirteen is signed this afternoon, with a surge on top of the largest range of the year.

WHAT YOU DECIDE 15W

Show the expansion is legitimate at the springs, and decide what the barrage closes on.

12.1 How far the expansion reaches

PREDICTION OFFICE · A ROOM · DERIVE

SCENE 37W

Tomorrow, the surge gets added in. The head will then move 1.40 metres around a mean of 2.1. Calloway won't carry the expansion into the springs prediction. Not until someone shows it still converges that far out.

GUIDE 84W

You build the working one line at a time. The test to use here compares each term with the one right before it. It's the ratio test. Form that ratio, simplify it, and see what it settles down to as the terms keep going. What it settles down to turns out to be a multiple of the movement. From there, the condition for the series to settle on a total comes in. It tells you exactly how large that movement is allowed to be.

ASK 14W

How large can the head's movement get before the expansion stops converging at all?

BEHIND THE BUTTON 187W

Why the ratio, and not just the size of a term. Each term here is a coefficient times a power of the movement. The coefficients alone don't obviously shrink or grow. But the ratio of one term to the next settles down quickly. That ratio is exactly what this test is built to use.

What the radius actually means, physically. The series converges as long as the movement stays smaller than the mean head. In other words, it converges as long as the head never actually reaches zero. That is not a coincidence of the math. At zero head, the three-halves power has an infinite derivative. No polynomial can follow a curve that steep.

Why this matters today and didn't matter yesterday. At 0.85 metres of movement, the ratio is comfortably small and nobody even asks the question. At tomorrow's 1.40 metres, including the surge, the ratio is two-thirds. Still inside the limit, but no longer comfortable. Suppose the surge had instead been forecast at 2.2 metres. Then the expansion could not have been used at all. The law would then have had to be computed directly.

TAKES AS READ

the terms of the expansion are a coefficient times a power of the movement

VERDICT 103W

The series converges as long as the movement stays smaller than the mean head. The ratio of consecutive terms is the movement divided by the head. Times a factor built from the two coefficients. That extra factor settles to one as the terms go on. So what's left, in the end, is just the movement over the head. The condition for the series to settle on a total is simple. This ratio must be less than one. Tomorrow's movement gives two-thirds, so the expansion is legitimate to use. That answers today's question — it says nothing about whether the surge itself is predictable.

TAKEAWAY 20W

A radius of convergence limits where a series may be used. It comes straight from the ratio of consecutive terms.

12.2 A hundred points of survey money

WALL STATION · A PERSON · SCIENCETANK

SCENE 36W

Grieve has a hundred points of survey money left this year. There are four things she could spend it on. The licence review happens in six weeks, and it will read whatever gets done this fortnight.

GUIDE 123W

Here are the four things the station could buy, and what it already has. The returned fraction of 0.62 came from a fortnight of gauge records. It was worked out from data, not measured straight off the wall. The shallow-water term is now built into the prediction. But it is fitted using six stations. Two of those have not been leveled since they were built. The stilling well at the outer face is the pipe that slows the gauge's reading. It needs forty minutes to settle. That is why the gate call cannot wait for a reading from it. Two flats stations need the survey launch and were missed last quarter. The wall itself was last surveyed before the eastern gate was rebuilt.

ASK 7W

Spread a hundred points across the four.

TAKES AS READ

money spent on a measurement is worth what the measurement would change

VERDICT 99W

The wall survey and the leveling both replace a number from records. They replace it with one that is actually measured. Both also feed decisions made again and again. These are the armour sizing, and every future prediction at those stations. The fast pressure sensor is worth something too. That's because the gate-timing argument depends on the stilling well needing forty minutes to settle. A sensor that skipped that wait would remove the trade-off, not just improve it. The launch charter is real work, but it only improves a record, not a decision. That is why it comes last.

TAKEAWAY 21W

Evidence is worth what it would change about a decision. It is not worth what it would add to a record.

12.3 What the gate closes on

SLUICE CONTROL · A ROOM · DIAGNOSIS

SCENE 36W

Oyelaran has four things on the desk and one signature to give. The programme goes to the gate crews at four o'clock, and the marsh warning with it. Tomorrow is the largest range of the year.

GUIDE 66W

Four readings of the same four pieces of paper. Look at every one of them before choosing, including the two that are not alarming. One is a bound that has been computed honestly. The other is a fit that now contains a term it did not have. What separates the explanations is which of the four things on the desk the level tomorrow actually depends on.

ASK 11W

Which account fits all four pieces of paper on the desk?

TAKES AS READ

a decision has to name the quantity it is being made on

VERDICT 101W

Three of the four pieces of paper are in good order. The fourth is different from the other three. The prediction is better than it has ever been and its bound is honestly computed. The shallow-water term is a real improvement fitted to real records. All of that is about terms that repeat. The surge does not repeat, so it is in none of those models and therefore in none of their bounds. It is half a metre on the largest range of the year. The gate closes on the gauge, with the prediction deciding when somebody is standing at it.

TAKEAWAY 20W

A prediction with a bound and a term added is still silent about anything that is not in the sum.

END OF THE DAY 22W

A prediction, a bound and a measurement are three different things, and a decision has to say which one it rests on.

THE CLOSING CARD

The eastern gate came down on the gauge, forty minutes later than the board would have had it, and the marsh stayed dry by about a foot. The springs ran to 11.6 m against a predicted 11.4, and the surge accounted for the difference almost exactly. The prediction the licence review will read now carries three things it did not carry a fortnight ago: the shallow-water term that the eight-constituent fit had been leaving in the residuals since commissioning, a bound of 0.19 m on the constituents nobody adds in, and a sentence saying that neither of those covers weather.

What it cost: one generation window, four hours of turbine time on the biggest ebb of the year, and an afternoon of the contractor being told no. What is unfinished: the returned fraction at the wall is still worked out of gauge records rather than measured off the wall, so the armour on the last forty metres is sized for a first arrival rather than for a total; two of the six stations feeding the shallow-water fit have not been levelled since the barrage was built; and the stilling well still needs forty minutes, which is the whole reason any of this had to be argued.

You took the prediction apart into the terms it is made of and put a ceiling on the ones nobody adds in. You found the term the residuals had been carrying for nine years, and you showed the expansion that produces it is still legitimate at the largest range of the year. And on the last afternoon you said out loud what neither bound covered, which is why the gate closed on a gauge reading and four hundred hectares of marsh had stock on it the next morning. Two signatures that had been in tension for nine years went on the same sheet. You wrote the paragraph that let them.

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